# Housing First or Treatment First?

The model, its parameters, its calibration, and what the referees still want

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**Abstract**

A complete statement of the rebuilt model: environment, preferences, budget, every transition, the household problem, and the equilibrium. Section 3 lists *every* parameter with its symbol, its role, and either its external source or the moment that pins it. Sections 4–7 cover the two-stage calibration, the untargeted moments used for validation, and the mechanism decomposition that replaces the static appendix. Section 8 lists, referee by referee, every comment that is still outstanding.

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# 1 Overview

The model is a life-cycle economy in which households choose housing, drug use, treatment, hours and saving under mortality, employment, mental-health and addiction risk, and in which a housing voucher programme is run under one of two rules. The question is whether the voucher should be conditioned on abstinence.

Relative to the submitted paper the model has been rebuilt rather than patched. Homeownership, the housing market, tenure, mortgages and negative assets are gone; the drug price is now an equilibrium object; treatment exists; mental illness exists; preferences no longer vary with addiction status; labour supply is continuous; and every parameter is attached either to a citation or to a moment. Table 8 in Section 8 maps each change to the comments it answers.

Two conventions used throughout. Time is annual. All monetary quantities are in units of average model income, so "0.25" means a quarter of average income.

# 2 The model

## 2.1 Demographics and timing

Households enter at model age $j = 1$ (real age 26) and live at most to $j = J = 60$ (real age 85). They retire at $j_R = 41$ (real age 66). Survival from $j$ to $j + 1$ is $\phi_j(h', d, m)$, defined in (13) below: it depends on age, on the housing chosen this period, on substance use disorder and on mental illness. A household that dies leaves its assets behind; the estate accrues to the government (Section 2.14).

Within a period the order of events is: the household observes its state, chooses hours, housing, drug use, treatment and saving, consumes, and then the shocks to productivity, employment, addiction, mental health, voucher receipt and survival are realised.

## 2.2 The state

Table 1: The individual state: 12,960 points per age, 777,600 in total

| | Symbol | Points | |
|---|---|---|---|
| Age | $j$ | 60 | real age 26–85, retirement at 66 |
| Financial assets | $a \geq 0$ | 60 | cubic spacing, dense near zero; no borrowing |
| Labour productivity | $y$ | 9 | Rouwenhorst discretisation of an AR(1) in logs |
| Employment | $e$ | 3 | employed; unemployed *with* UI; unemployed without UI |
| Substance use disorder | $d$ | 2 | $d = 0$ clean, $d = 1$ SUD |
| Serious mental illness | $m$ | 2 | $m = 0$ absent, $m = 1$ present |
| Housing voucher | $\eta$ | 2 | $\eta = 0$ not held, $\eta = 1$ held |

Two features are deliberate.

**Housing entered with is not a state.** Survival, employment transitions and the recovery hazard all condition on the *chosen* housing $h'$, and renters face no adjustment cost, so the unit a household arrives in is payoff-irrelevant. Carrying it would replicate the whole problem $N_h$ times; removing it leaves every aggregate unchanged and cuts the solve by a factor of 6.6. It would have to be reinstated if moving costs were added.

**Three employment states, not four.** UI is drawn on *entry* into unemployment with probability $\pi_{UI}$ and lasts one year. A first-year non-recipient and a household unemployed for two or more years have identical income (zero) and identical continuation values, so those two states merge *exactly*. This answers the AE's A2: UI expiration matters, and it costs one state rather than two.

### 2.3 Choices

Each period a household chooses

$$\big(n,\ h',\ v,\ \tau,\ a'\big) \in [0,1] \times \{h_1, \ldots, h_4\} \times \mathbb{R}_+ \times \{0,1\} \times \mathcal{A},$$

hours $n$, housing $h'$ (where $h_1$ denotes homelessness and $h_2 < h_3 < h_4$ are rental sizes), drug consumption $v$, treatment take-up $\tau$ (available only if $d = 1$), and next-period assets $a'$ on the asset grid.

### 2.4 Preferences

Period utility is

$$u = \frac{\left[c^{\alpha}\,(h')^{\beta}\,(v + b_{dmh'})^{\gamma}\right]^{1-\sigma}}{1-\sigma} \;+\; \omega_{\ell}\frac{(1-n)^{1-\sigma_{\ell}}}{1-\sigma_{\ell}} \;+\; \aleph \;-\; \tau\,\bar{\tau}^{u} \;-\; \kappa_x\big(H + \omega_x H_D\big), \tag{1}$$

with the Cobb–Douglas weights

$$\alpha = (1-\theta_h)(1-\theta_v), \qquad \beta = \theta_h(1-\theta_v), \qquad \gamma = \theta_v, \qquad \alpha+\beta+\gamma = 1.$$

**The weights do not depend on addiction status.** Referee 2 (C5), Referee 3 (M2a) and the AE (A8) all objected that in the submitted paper the taste weight on drugs rose with $d$, so reducing SUD mechanically raised the utility derived from $c$ and $h$ and made addiction-reduction and housing expansion complements "even absent incentive effects". Fixing $\alpha, \beta, \gamma$ removes that channel completely. It is also a *commitment*: with preferences invariant, becoming addicted no longer relaxes any constraint, so every private cost of SUD must now come from mortality, employment and the budget. Section 7 tests exactly that.

**Only the Stone–Geary shifter varies.** The AE's suggested alternative (A8) was to vary $b$ instead, and that is what the model does:

$$b_{dmh'} = b_d \;\cdot\; \rho_m^{\mathbf{1}\{m=1\}} \;\cdot\; \rho_H^{\mathbf{1}\{h'=h_1\}}, \qquad d \in \{0,1\}, \quad \rho_m, \rho_H \in (0,1]. \tag{2}$$

A lower $b$ raises the marginal utility of the first units of use. Three separate things are going on here and each is doing a job.

1. $b_0$ *versus* $b_1$. A single $b$ cannot match both the participation rate and the drug-expenditure share. A $b$ large enough to deter clean households pushes the unconstrained interior choice $v^*$ below zero, which drives expenditure by households *with* SUD to the corner. Two values separate the moments: $b_0$ pins participation, $b_1$ keeps the choice interior so that $\theta_v$ pins the expenditure share.

2. $\rho_m < 1$*: self-medication.* Poor mental health raises the marginal value of the first units of use. This is the channel Referee 1 asks about in comment 1(a).

3. $\rho_H < 1$*: the housing-access channel.* This is the only parameter in the model that governs the elasticity of drug use with respect to housing access — the object Referee 2 (C1) calls the crux on which all the welfare results rest, and which Referee 3 and the AE (A13) name independently. Without it that moment is not merely untargeted, it is *untargetable*: housing a household only raises its resources, drugs are a normal good, and the model would predict that housing *raises* use, the opposite sign to Milby et al. (2005). Section 4.2 calibrates $\rho_H$ to that trial.

**Leisure.** $\omega_\ell$ is the weight on leisure and $\sigma_\ell$ its curvature, which pins the Frisch elasticity at $1/\sigma_\ell$.

**ℵ, the flow value of being alive.** ℵ is what makes mortality risk deter drug use: without it, dying is not costly and the model's central mechanism disappears. It also has to satisfy a condition that is easy to overlook — see Section 6.

**$\bar{\tau}^u$, the utility cost of treatment.** Treatment take-up costs utility, not resources. A resource cost comes out of the same budget that buys food and rent, so the households with the strongest case for treatment — the poorest — would be exactly the ones unable to afford it, which is not how publicly funded treatment works. $\bar{\tau}^u$ is the stigma, effort and time of entering a programme whose monetary cost somebody else bears.

**The externality.** $H$ is the aggregate homelessness rate and $H_D$ the share of the population both homeless and with SUD. Referee 2's wording is specific — "a large homeless population, *especially if many are addicted to drugs*, creates a negative externality beyond the fiscal burden" — so the disamenity is weighted by composition through $\omega_x$ rather than depending on the headcount alone. Households take $H$ and $H_D$ as given, which is what makes this a spillover rather than a private cost, but both are equilibrium objects, so policy moves them and the movement feeds back through the value of life into drug decisions. That is the property $\kappa_{\text{cons}}$ lacked and which Referee 1 attacked in comment 3(a). The baseline sets $\kappa_x = 0$; see Section 7.6.

### 2.5 Income, taxes and transfers

Gross income depends on age, employment, productivity and health:

$$y^g(j,y,e,d,m,n) = \begin{cases} 0.40\,y & j \geq j_R \quad \text{(social security)} \\ n\,y\,\varepsilon_j\,\rho_d^{\mathbf{1}\{d=1\}}\rho_m^{\mathbf{1}\{m=1\}} & j<j_R,\ e=1 \quad \text{(employed)} \\ \varrho_{UI}\,y\,\varepsilon_j & j<j_R,\ e=2 \quad \text{(UI, first year only)} \\ \zeta_{DI}\,\bar{y}_{DI}\,\bar{Y} & j<j_R,\ e=3,\ m=1 \quad \text{(disability)} \\ 0 & \text{otherwise,} \end{cases} \tag{3}$$

where $\varepsilon_j$ is the deterministic age profile of efficiency, $\rho_d$ and $\rho_m$ are productivity penalties *conditional on working*, and $\bar{Y}$ is average income.

**Why the penalties are conditional on working.** This is what allows an *employed* household to have income low enough to choose homelessness. An unemployment effect alone cannot do that, and Referee 1's comment 2(c)(i) is precisely that the model produced a homeless population that was 100% out of work against 40–52% in the data.

Labour income is taxed with the Heathcote–Storesletten–Violante function and capital income at a flat rate:

$$y^n = \tau_\ell\,(y^g)^{\xi} \;+\; (1-\tau_k)\,r\,a. \tag{4}$$

Two transfers. A *food-stamp floor* tops means-tested income up to $\underline{c}\,\bar{Y}$:

$$F = \max\Big\{0,\ \underline{c}\,\bar{Y} - \big(y^g + \max\{0, ra\}\big)\Big\}, \tag{5}$$

and a *housing voucher*, paid only to voucher holders and only while renting, covering rent above a tenant contribution of $\kappa_r$ of gross earnings:

$$Vch = \mathbf{1}\{\eta=1\}\mathbf{1}\{h' \neq h_1\}\max\big\{0,\ p_h h' - \kappa_r\,y^g\big\}. \tag{6}$$

Both are means-tested on *gross* income, as in the programmes themselves. Both are also clawed back — at a rate of one for food stamps and $\kappa_r$ for the voucher — and that is what generates the welfare trap discussed under labour supply below.

### 2.6 The budget constraint

$$c + p_v v + a' \;=\; \underbrace{y^n + a}_{\text{after-tax income and assets}} \;-\; \underbrace{p_h h' \mathbf{1}\{h' \neq h_1\}}_{\text{rent}} \;+\; Vch + F, \qquad c>0,\ \ v\geq 0,\ \ a'\geq 0. \tag{7}$$

Note what is *not* in (7): treatment, which costs utility; and bequests, which go to the government.

**The closed-form split of $c$ and $v$.** Given $a'$ and $h'$, let $R$ denote the resources remaining. Cobb–Douglas preferences over $(c,\ v+b)$ give

$$c = \frac{\alpha}{\alpha+\gamma}\big(R + p_v b\big), \qquad v = \max\Big\{0,\ \frac{\gamma}{(\alpha+\gamma)p_v}\big(R+p_v b\big) - b\Big\}, \tag{8}$$

so the inner allocation never needs a numerical search. The max is the corner at which the household abstains because the Stone–Geary endowment already exceeds its desired intake — which is exactly how a large $b$ deters use.

### 2.7 Labour supply

Hours are continuous and satisfy the intratemporal first-order condition

$$u_c(c)\,\frac{\partial R}{\partial n} \;=\; \omega_\ell (1-n)^{-\sigma_\ell}. \tag{9}$$

Two properties of (9) matter substantively.

**The residual is not monotone: the welfare trap.** From (5) and (6), $\partial R/\partial n$ jumps *up* at the hours where each transfer is exhausted, because the clawback stops. Below the food-stamp exhaustion point the marginal return to an hour of work is close to zero. So "work a little and keep the transfer" and "work properly" are genuinely distinct local optima, and the household compares them. The solution algorithm splits $[0,1]$ at the kinks and solves each piece exactly; bracketing once would silently pick whichever optimum happened to be bracketed.

**Eligibility is a step in the continuation value.** Voucher eligibility is lost discretely when means-tested income crosses $\kappa_V$ times the median. That does not kink the budget — so (9) is continuous across it — but it shifts the continuation value, so the piecewise optimum is clamped to each side of the threshold and both are evaluated.

Referee 3 (m1) wanted labour supply kept and its role made visible; the AE (A3) questioned whether it was needed. Making it continuous and exact resolves the tension in Referee 3's favour, at a cost of a factor of three in solve time.

### 2.8 Housing and homelessness

Housing is a four-point grid. $h_1$ is homelessness, with a small service flow $h_{\mathrm{HL}}$ and no rent. The three rental sizes are geometric,

$$h_2 = \underline{h}, \qquad h_3 = \underline{h}\,\varsigma, \qquad h_4 = \underline{h}\,\varsigma^2, \tag{10}$$

so two parameters describe the whole grid: the level $\underline{h}$ and the spread $\varsigma$.

**Rent is normalised.** Only the product $p_h h$ enters both the budget (7) and utility (1), so the rent level and the housing grid are not separately identified. Setting $p_h = 1$ makes the *grid* the calibrated object, and $\underline{h}$, $\varsigma$ and $h_{\mathrm{HL}}$ are then pinned by three moments of the housing-burden distribution and the homelessness rate. This is what Referee 2's comment D1 asked for: in the submitted paper the grid had seven points, four of them uncalibrated and listed in a footnote.

### 2.9 Drug use, addiction, recovery and treatment

A household either abstains ($v = 0$) or uses ($v > 0$, chosen continuously according to (8)). Transitions in $d$:

$$\Pr(d' = 1 \mid d = 0) = \pi_d \cdot \mathbf{1}\{v > 0\}, \qquad \Pr(d' = 0 \mid d = 1) = \pi_r(h', \tau, m). \tag{11}$$

**A single addiction hazard.** A clean household that uses becomes addicted with probability $\pi_d$ *regardless of how much it consumes.* There is no light/heavy threshold $\bar{v}$ and no consumption-dependent hazard; the 6.48 heavy-to-light ratio is gone. This removes two of the parameters Referee 3 (M2b) asked to have justified, at the cost of one channel: intensity no longer affects the risk of addiction, only the budget.

**Recovery and treatment.** The recovery hazard is

$$\pi_r(h', \tau, m) = \min\Big\{1,\ \bar{\pi}_r \cdot \mu_H^{\mathbf{1}\{h'=h_1\}} \cdot \mu_m^{\mathbf{1}\{m=1\}} \cdot g^{\tau}\Big\}, \tag{12}$$

with $\mu_H < 1$ (recovery is harder when homeless), $\mu_m < 1$ (harder with mental illness) and $g > 1$ the treatment effect. $\tau \in \{0, 1\}$ is a choice, available only to households with $d = 1$, at the utility cost $\bar{\tau}^u$ in (1).

Referee 1's comment 1(b) and the AE's A10 are the same point and it is the most damning gap in the submitted paper: the policy is called Treatment First and there was no treatment in the model. There is now, it is a decision rather than a mechanical hazard, and $g$ — through which the entire Treatment First result runs — is disciplined in Section 4.2.

**The maximum experimentation age.** In the baseline a clean household may initiate only up to $j \leq j_E$. Referee 1's smaller comment 1 notes this is in tension with the U-shaped age profile of use in Table 4(a). It is now a variable rather than a hard-coded parameter, so the model can be run without it; Section 6 reports the age profile both ways.

## 2.10 The private costs of SUD: mortality, employment, and the homelessness loop

This subsection is the heart of the mechanism and it deserves to be stated explicitly rather than buried in a parameter table.

**Mortality.** Survival is

$$\phi_j(h', d, m) = 1 - \min\Big\{1,\ \big(1 - \bar{\phi}_j\big) \cdot \kappa_{\phi H}^{\mathbf{1}\{h'=h_1\}} \cdot \kappa_{\phi d}^{\mathbf{1}\{d=1\}} \cdot \kappa_{\phi m}^{\mathbf{1}\{m=1\}}\Big\}, \tag{13}$$

where $\bar{\phi}_j$ is the baseline age profile of survival. SUD multiplies the death hazard by $\kappa_{\phi d}$, homelessness by $\kappa_{\phi H}$ and mental illness by $\kappa_{\phi m}$. Because $\aleph > 0$, a higher death hazard is a real cost, and it is larger for households with more remaining life and more to consume — which is the sorting the paper's mechanism relies on.

**Employment.** Job finding and separation are

$$\pi^{ue} = \min\big\{1,\ \bar{\pi}^{ue}\, \kappa_{ueH}^{\mathbf{1}\{h'=h_1\}} \kappa_{ued}^{\mathbf{1}\{d=1\}} \kappa_{uem}^{\mathbf{1}\{m=1\}}\big\}, \qquad \pi^{eu} = \min\big\{1,\ \bar{\pi}^{eu}\, \kappa_{ueH}^{-\mathbf{1}\{h'=h_1\}} \kappa_{ued}^{-\mathbf{1}\{d=1\}} \kappa_{uem}^{-\mathbf{1}\{m=1\}}\big\}, \tag{14}$$

so each multiplier below one simultaneously makes work harder to find and easier to lose. On separation the household draws UI with probability $\pi_{UI}$:

$$e = 1 \ \longrightarrow\ \begin{cases} e' = 1 & \text{w.p. } 1 - \pi^{eu} \\ e' = 2 & \text{w.p. } \pi^{eu}\pi_{UI} \\ e' = 3 & \text{w.p. } \pi^{eu}(1 - \pi_{UI}) \end{cases} \qquad e \in \{2, 3\} \ \longrightarrow\ \begin{cases} e' = 1 & \text{w.p. } \pi^{ue} \\ e' = 3 & \text{w.p. } 1 - \pi^{ue}. \end{cases}$$

**The homelessness loop.** Homelessness enters (13), (14), (12) and (2): it raises mortality, worsens employment, lowers the recovery hazard, and raises the marginal utility of drug use. Those four together are what make homelessness and SUD mutually reinforcing, and shutting them off jointly is arm 4 of the mechanism ladder.

**What SUD does *not* do.** In the baseline there is no direct wage penalty conditional on working, $\rho_d = 1$. The AE (A20) asks about direct productivity effects. The robust findings are on the employment margin — roughly 50% more unscheduled absence and 44% higher turnover, captured by $\kappa_{ued}$ — while the wage penalty conditional on working is contested, with parts of the older literature finding positive effects for casual use. The baseline therefore imposes none and says so; a version calibrated to the SUD income ratio is reported as an alternative.

### 2.11 Serious mental illness

$m$ is an exogenous two-state Markov chain. It represents *serious* mental illness, not any mental illness: AMI covers roughly 23% of US adults and includes mild anxiety and depression, whereas SMI is about 5.5% and is defined by substantial functional impairment. A state that halves productivity cannot sensibly cover a quarter of the population.

The onset rate is *derived* from the target prevalence $p_m$ and the recovery rate $\lambda_m$,

$$\text{onset} = \lambda_m \frac{p_m}{1 - p_m}, \tag{15}$$

so calibrating persistence never disturbs prevalence.

SMI affects six objects: productivity conditional on working ($\rho_m$), job finding and separation ($\kappa_{uem}$), mortality ($\kappa_{\phi m}$), the recovery hazard ($\mu_m$), the Stone–Geary shifter ($\rho_m$ in (2)), and eligibility for disability income ($\zeta_{DI}$ in (3)).

The specification follows İmrohoroğlu and Zhao (2022) structurally, not just numerically: in their model labour-market shocks generate *short* homeless spells while health shocks generate *chronic* homelessness, and expanded means-tested disability income is the effective policy for the chronic group. Homelessness driven purely by income risk is transitory; the persistent tail needs a persistent state. This is Referee 1's comment 1(a), and the persistence is also what generates the older homeless population of comment 2(c)(ii).

### 2.12 The voucher programme and the two policies

A household without a voucher draws one with probability $\pi_V$ if it is income eligible; a household with one keeps it while eligible. Eligibility is

$$y^g + \max\{0, ra\} \;<\; \kappa_V \cdot \text{median income}. \tag{16}$$

Under **Housing First** that is the whole rule. Under **Treatment First** a voucher is additionally lost with probability $p_{\text{det}}$ if the household used drugs, and with probability $p_{\text{fp}}$ even if it did not:

$$\Pr(\eta' = 1) = \Pr(\eta' = 1)^{HF} \times \begin{cases} 1 - p_{\text{det}} & \text{if } v > 0 \\ 1 - p_{\text{fp}} & \text{if } v = 0. \end{cases} \tag{17}$$

Referee 1 (4a, 4b) and the AE (A11) both note that perfect observation and enforcement are not credible. $p_{\text{det}}$ and $p_{\text{fp}}$ are experiment parameters reported as sensitivity, not calibrated, because that is the point.

### 2.13 The household problem

Let $s = (a, y, e, d, m, \eta)$. The value function solves

$$V_j(s) = \max_{n,h',v,\tau,a'} \frac{\left[c^{\alpha}(h')^{\beta}(v + b_{dmh'})^{\gamma}\right]^{1-\sigma}}{1-\sigma} + \omega_{\ell}\frac{(1-n)^{1-\sigma_{\ell}}}{1-\sigma_{\ell}} + \aleph - \tau\bar{\tau}^{u} - \kappa_x\big(H + \omega_x H_D\big) + \beta\,\phi_j(h', d, m)\,\mathbb{E}\Big[V_{j+1}(a', y', e', d', m', \eta') \;\Big|\; s, h', v, \tau\Big] \tag{18}$$

subject to (3)–(7), with $V_{J+1} \equiv 0$: there is no bequest motive, and death is worth zero. The expectation is taken over the independent transitions in $y$ (Rouwenhorst), $e$ (14), $d$ (11), $m$, and $\eta$.

The problem is solved by exact backward induction. Within a drug regime and a housing choice the $c$–$v$ split is closed form (8) and hours come from (9), so the only search is over $a'$, done on the full grid because the objective is *not* single-peaked in $a'$: voucher eligibility is a step function of income (which includes $ra$), the food-stamp floor kinks, and housing is discrete.

### 2.14 Government

The government pays social security, UI, disability, food stamps and vouchers, and bears a fiscal cost $\chi_H$ per homeless household and $\chi_D$ per household with SUD. It finances this with the labour income tax (4), the capital income tax, and the estates of the deceased. The level parameter $\tau_{\ell}$ clears the budget:

$$\underbrace{\sum y^{g} - \tau_{\ell}\sum (y^{g})^{\xi} + \tau_k \sum \max\{0, ra\} + \text{estates}}_{\text{revenue}} = \underbrace{SS + UI + DI + F + Vch + \chi_H H + \chi_D D + G}_{\text{outlay}}. \tag{19}$$

**Estates go to the government.** Redistributing accidental bequests as a uniform lump sum is a large unconditional transfer to the poor with welfare consequences of its own, which would contaminate every policy comparison in the paper. Sending them to the budget also removes an equilibrium fixed point. The lump-sum version remains available for a robustness run. This is a change from the submitted paper, where transfers were additionally type-dependent — the AE's A7.

### 2.15 Drug supply and the price

Drug production has cost $C(V) = \frac{\psi_1}{\psi_2}V^{\psi_2}$, so supply satisfies $p_v = \psi_1 V^{\psi_2 - 1}$ and the price elasticity of supply is $1/(\psi_2 - 1)$. The shifter $\psi_1$ is normalised so that $p_v = 1$ at the benchmark quantity.

This answers Referee 1 (1c), Referee 3 (M2d) and the AE (A5). Referee 3 worried that endogenising the price "requires suppressing this normalization". It does not: the normalisation moves to $\psi_1$, the benchmark price is still one, and only counterfactuals move it.

**Honest statement about $\psi_2$.** There is no credible estimate of the supply elasticity of illicit drugs. It is reported as a bounded sensitivity, and note that the submitted paper's fixed price is the $\psi_2 \to 1$ endpoint of that range, so the sensitivity brackets the old assumption rather than replacing it with a different guess.

### 2.16 Stationary equilibrium

A stationary equilibrium is a value function, policy functions, a distribution $\Phi$, and $(\tau_\ell, p_v, \bar{Y}, \text{median}, H, H_D)$ such that households solve (18); $\Phi$ is the invariant distribution induced by the policies and the transitions; the government budget (19) holds; the drug market clears at $p_v$; and $\bar{Y}$, the median, $H$ and $H_D$ are the aggregates implied by $\Phi$.

Note that $H$ and $H_D$ enter (1), so they are genuine fixed points whenever $\kappa_x > 0$, and that the median enters voucher eligibility, so it is a fixed point too — in a skewed income distribution the median sits well below the mean, and using the mean would materially overstate the eligible population.

## 3 Parameters

Every parameter in the model. **Status** is one of: **N** normalisation or definition; **L** standard in the literature, cited; **D** set directly from data; **C** internally calibrated, with the moment it is assigned to; **E** determined in equilibrium; **S** reported as a sensitivity rather than calibrated, because no credible point estimate exists.

Table 2: Preferences and prices

| Symbol | Meaning | St. | Value, source or target |
|---|---|---|---|
| $\sigma$ | risk aversion | L | 2.0, standard |
| $\sigma_\ell$ | curvature on leisure | L | 3.0; Frisch elasticity 0.5, Chetty (2012) |
| $\omega_\ell$ | weight on leisure | C | $\to$ average hours worked, 40% of time |
| $\theta_h$ | Cobb–Douglas weight on housing | C | $\to$ aggregate housing cost / income, 21.2% |
| $\theta_v$ | Cobb–Douglas weight on drugs | C | $\to$ drug spending / income given SUD, 44% |
| $b_0$ | Stone–Geary shifter, clean | C | $\to$ drug participation, 5.13% |
| $b_1$ | Stone–Geary shifter, SUD | C | keeps $v$ interior so $\theta_v$ identifies the expenditure share |
| $\rho_m$ | $b$ multiplier, mental illness | C | $\to$ SUD rate given SMI, 9.75% |
| $\rho_H$ | $b$ multiplier, homelessness | $C^2$ | $\to$ Milby housed/unhoused abstinence ratio, 1.473 |
| $\aleph$ | flow value of being alive | C | $\to$ SUD income / average, 68.7%; model VSL reported against \$7.5m |

| Symbol | Meaning | St. | Value, source or target |
|---|---|---|---|
| $\bar{\tau}^u$ | utility cost of treatment | C | $\rightarrow$ treatment take-up given SUD, $\approx$24% |
| $\beta$ | discount factor | C | $\rightarrow$ wealth / GDP = 3 |
| $p_h$ | rent | N | = 1; only $p_h h$ is identified |
| $p_v$ | drug price | E | clears the drug market; = 1 at the benchmark |
| $r$ | risk-free rate | L | 0.03, Kaplan, Mitman and Violante (2020) |
| $\kappa_x$ | externality weight | S | 0 in the baseline; see §7.6 |
| $\omega_x$ | extra weight on homeless-with-SUD | S | 2.0, with $\kappa_x$ |

Table 3: Housing, taxes and transfers

| Symbol | Meaning | St. | Value, source or target |
|---|---|---|---|
| $h_{\text{HL}}$ | service flow of homelessness | C | $\rightarrow$ homelessness rate, 0.30% |
| $\underline{h}$ | smallest rental size | C | $\rightarrow$ share with housing cost > 50% of income, 17.49% |
| $\varsigma$ | geometric spread of rental sizes | C | $\rightarrow$ share with housing cost > 30% of income, 40.62% |
| $\xi$ | tax progressivity (HSV) | L | 0.85, Heathcote, Storesletten and Violante (2017) |
| $\tau_\ell$ | tax level | E | clears the government budget |
| $\tau_k$ | capital income tax | L | 0.15 |
| $\underline{c}$ | food-stamp consumption floor | C | $\rightarrow$ food stamps / labour income, 0.42% (SNAP outlays FY2023) |
| $\kappa_V$ | voucher eligibility threshold | D | 0.50 of median income, HUD rule |
| $\kappa_r$ | tenant contribution rate | D | 0.30, HUD rule |
| $\pi_V$ | voucher lottery probability | C | $\rightarrow$ voucher recipients / income-eligible, 26.6% (HUD) |
| $p_{\text{det}}$ | detection probability under TF | S | 1.0 in the benchmark; swept |
| $p_{\text{fp}}$ | false-positive rate under TF | S | 0.0 in the benchmark; swept |
| $\chi_H$ | fiscal cost per homeless household | D/S | 0.252; National Alliance to End Homelessness. R2 and R3 both judge it high — run as sensitivity against Abramson and van Nieuwerburgh |
| $\chi_D$ | fiscal cost per household with SUD | D | 0.212; CEA (2017), $\approx$\$30,000 per person-year |

| Symbol | Meaning | St. | Value, source or target |
|---|---|---|---|
| $G$ | other public spending | E | residual, then held fixed |

Table 4: Income process, employment and mortality

| Symbol | Meaning | St. | Value, source or target |
|---|---|---|---|
| $\rho_y, \sigma_y$ | AR(1) in log productivity | L | 0.90, 0.20 |
| $\varepsilon_j$ | deterministic age profile | D | quadratic in age |
| $\bar{\phi}_j$ | baseline survival | D | **placeholder**; replace with US life tables |
| $\bar{\pi}^{ue}$ | job finding, annual | L | 0.55, Krueger, Cramer and Cho (2014) |
| $\bar{\pi}^{eu}$ | job separation | C | $\rightarrow$ unemployment rate, 4.9% |
| $\pi_{UI}$ | share of entrants drawing UI | D | 0.30; insured unemployment / total unemployment, FRED 2014–24 |
| $\varrho_{UI}$ | UI replacement rate | D | 0.50, standard US |
| $\kappa_{ueH}$ | employment multiplier, homeless | L | 0.80, Desmond and Gershenson (2016) |
| $\kappa_{ued}$ | employment multiplier, SUD | C | $\rightarrow$ unemployment rate given SUD, 26.9% |
| $\kappa_{uem}$ | employment multiplier, SMI | C | $\rightarrow$ unemployment rate given SMI, 7.1% |
| $\kappa_{\phi H}$ | mortality multiplier, homeless | D | 1.60; Meyer et al. (2025), 60% excess mortality |
| $\kappa_{\phi d}$ | mortality multiplier, SUD | $C^2$ | $\rightarrow$ drug-price participation elasticity, $-0.63$; bracketed by the 2.5–9.8 SMR range of Degenhardt et al. (2011) |
| $\kappa_{\phi m}$ | mortality multiplier, SMI | D | 2.50; standardised mortality ratio for SMI of 2–3 |
| $\rho_d$ | SUD productivity penalty | D | 1.00 in the baseline: no direct wage effect (AE A20) |
| $\rho_m$ | SMI productivity penalty | C | $\rightarrow$ homeless income / average, 19.1% |

Table 5: Addiction, recovery, treatment and mental illness

| Symbol | Meaning | St. | Value, source or target |
|---|---|---|---|
| $\pi_d$ | addiction hazard given any use | C | $\rightarrow$ SUD rate, 3.68% |

| Symbol | Meaning | St. | Value, source or target |
|---|---|---|---|
| $j_E$ | maximum experimentation age | S | real age 55; run with and without (§6) |
| $\bar{\pi}_r$ | baseline recovery hazard, housed | D | 0.025/yr; Vaillant (1995). The AE questions applying alcohol recovery rates to drug SUD |
| $\mu_H$ | recovery multiplier, homeless | C | → drug UD share of the homeless, 21.7% |
| $\mu_m$ | recovery multiplier, SMI | S | 0.50; direction certain, magnitude not sourced |
| $g$ | treatment effect on recovery | S | 1.35, from an odds ratio of 1.22 for contingency management. **The TF result runs through this** — report a range and the value at which the ranking flips |
| $p_m$ | SMI prevalence | D | 5.5%, NSDUH 2023 |
| $\lambda_m$ | SMI recovery rate | D | 0.067/yr; İmrohoroğlu–Zhao's own moment, ≈50% improve enough to work within 10 years |
| $\text{onset}_m$ | SMI onset | N | derived from $p_m$ and $\lambda_m$ |
| $\zeta_{DI}$ | disability coverage given SMI | C | → SMI share of the homeless, 25% |
| $\bar{y}_{DI}$ | disability benefit level | S | 0.25 of average income; order of magnitude only |
| $\psi_1$ | drug supply shifter | N | $p_v = 1$ at the benchmark |
| $\psi_2$ | drug supply curvature | S | elasticity $1/(\psi_2 - 1)$; no credible estimate, swept |

**Superscript 2** marks the two parameters calibrated in the *second* stage, against experimental rather than cross-sectional moments.

**Parameters that remain undisciplined.** Five, and they are flagged rather than hidden: $g$ (treatment effect), $\mu_m$, $\bar{y}_{DI}$, $\psi_2$ and $\kappa_x$. Of these only $g$ carries a headline result, and it is reported as a range. The baseline survival profile $\bar{\phi}_j$ is currently a smooth functional form and should be replaced with US life tables before the paper is resubmitted.

# 4 Calibration

Two stages, and they are different exercises. The first matches a cross-section of the stationary equilibrium. The second matches short-run *responses* measured in experiments, which cannot be computed by comparing stationary equilibria.

### 4.1 First stage: the stationary equilibrium

Nineteen parameters are calibrated jointly against twenty-one moments of the stationary distribution. All the assignments are in Section 3; what matters methodologically is the following.

**The parameter–moment map is close to triangular, and the code says so.** Each parameter has one moment it dominates, and the sign of $\partial M/\partial\theta$ is declared explicitly for each pair, so a mistaken sign shows up immediately as a moment walking away from its target rather than as a silent failure.

**Three moments are step-like and must be bracketed, not stepped.** Participation is a near-threshold function of $b_0$ — below a cutoff everyone uses, above it nobody does, and in between every drug moment reads zero. Homelessness moved from 0.12% to 3.29% for a 35% change in $h_{\mathrm{HL}}$. Wealth/GDP is 5.46 at $\beta = 0.995$ and 0.38 at $\beta = 0.85$. A proportional step of 0.9 takes a discount factor from 0.96 to 0.864, which is enormous in a life-cycle model; the first attempt at this calibration drove $\beta$ into its lower bound and took every other moment with it. All three are bisected at the head of each sweep, in partial equilibrium, before anything else moves.

**The discount factor and wealth/GDP.** $\beta$ targets aggregate wealth equal to three times GDP, with GDP measured as *all factor income* — labour earnings plus the return on the capital stock — and transfers excluded, because transfers are redistribution rather than output. Three is the standard US ratio of the private capital stock to annual output. It is the right target here rather than the SCF's net-worth-to-*household-income* ratio of roughly 7.3, whose numerator is mostly housing equity a model without homeownership cannot hold and whose denominator is not output. One implication should be stated in the paper rather than left for a reader to find: with $A = 3 \cdot \mathrm{GDP}$ and $\mathrm{GDP} = \text{labour income} + rA$, capital income is 9% of GDP, low against a one-third capital share. That follows from holding $r$ fixed at 3% in a model with no production function.

**Three targets that were wrong in the submitted paper.**

1. *SUD among the homeless.* The paper targets 37%. Gutwinski et al. (2021) pool 39 studies of homeless people in high-income countries and report 36.7% *alcohol* use disorder and 21.7% *drug* use disorder. The model is about drug use. This is the AE's A16.

2. *Income of drug users.* Referee 1 (2c-i) attacked the model's 41% against "roughly 74%". The 74% is computed over people with a use *disorder*, not people who use. For ages 26+, NSDUH gives 95.1% for users and 68.7% for moderate or severe use disorder. Targeting the disorder ratio and *validating* on the use ratio is the sharper test, and it makes the economics explicit: use is close to income-neutral, and it is progression to disorder that concentrates at the bottom.

3. *Housing burden.* Everyone rents in this model, so a renters-only target would import the selection into who rents. The target is the aggregate ratio over *all* households with owner housing valued at imputed rent, and the full distribution is used for validation.

### 4.2 Second stage: short-run experimental moments

Referee 2 (C1): "all of the welfare results rest on whether the model has an accurate elasticity of drug use to housing access". Referee 3 and the AE (A13) say the same independently. Referee 1 (2d, 2e) asks for policy-relevant elasticities from the RCT literature and notes that the price-elasticity moment as written is not replicable.

These are short-run objects. A trial hands a treatment to a selected group and measures them months later; it does not move aggregate prices and it does not wait for a new stationary distribution. The second stage holds prices at benchmark, takes the benchmark distribution as the initial condition, restricts to the trial's eligible population, applies each arm's rules, re-solves the household problem, and pushes the cohort forward.

**The housing-access elasticity.** Milby et al. (2005) randomised 196 cocaine-dependent homeless adults into three arms, all receiving identical day treatment:

| Arm | Model counterpart | $n$ | Weeks abstinent (of 24) |
|---|---|---|---|
| NH | no assistance | 66 | 4.48 |
| NACH | voucher, unconditional — *Housing First* | 67 | 6.60 |
| ACH | voucher, conditional — *Treatment First* | 63 | 8.32 |

The targeted moment is the *ratio* NACH/NH = 1.473, which identifies $\rho_H$. Ratios rather than levels: "mean consecutive weeks abstinent out of 24" is not the same object as "abstains this model year", but the treatment effect is comparable and it is the treatment effect the referees want reproduced.

**The conditionality effect is a validation check, not a target.** The abstract reports that the abstinence-contingent arm beat the unconditional arm, but only *after controlling for treatment attendance*. In the raw randomised comparison NACH versus ACH gives $P = .091$, and the GEE models state the two housed groups "did not differ from one another". Calibrating to a point estimate the source reports as insignificant is exactly the error Referee 1 attacked $\kappa_{\text{cons}}$ for.

**The extensive/intensive contrast is the sharper test.** Milby's counterfactual is literally no housing, and the effect is large. The Housing First RCTs compare Housing First to "usual services", which already include shelter or transitional housing, and the systematic review finds "no clear differences ... on outcomes related to mental health, quality of life and substance use". These are not in conflict: the elasticity is *large at the extensive margin* and *near zero at the intensive margin*. A model reproducing both is far better identified than one matching either alone.

**The drug-price participation elasticity.** Perturb $p_v$, hold everything else at benchmark, re-solve, read the arc elasticity of participation. Partial equilibrium deliberately, since the empirical estimates come from price variation rather than from a comparison of steady states. Target $-0.63$; identifies $\kappa_{\phi d}$ within the Degenhardt SMR range. Referee 1 also asks for robustness across $-0.36$ to $-0.9$, which is the same bisection at the two endpoints.

**The value of a statistical life.** Reported rather than targeted, because ℵ currently carries a first-stage moment. Reference $7.5m (Viscusi and Aldy 2003; EPA 2010). If the model-implied value is far off, the right response is to move ℵ to the VSL target and reassign the SUD income ratio to $\rho_d$.

# 5 Validation

None of these is targeted. Referee 1's 2(a) — model fit never reported — and the AE's A12 are answered by reporting all of them, including the misses, and by framing the misses honestly rather than favourably (2b).

Table 6: Untargeted moments

| Moment | Data | Source |
|---|---|---|
| *Distributional* | | |
| Full CDF of housing cost / income | 12 thresholds, 5–100% | SCF 2022, ages 26–85, owner housing at imputed rent |
| Income of drug *users* / average | 95.1% | NSDUH 2024, ages 26+ |
| Income of those with SUD / average | 68.7% (targeted; reported alongside) | NSDUH 2024, ages 26+ |
| Share of the homeless employed | 40–52% | R1 comment 2(c)(i) |
| Age distribution of the homeless | 48.1% aged 26–44; 27.1% 55+ | R1 comment 2(c)(ii); to be reported in finer bins |
| Drug use by age | U-shaped | NSDUH microdata |
| Wealth by age | — | SCF 2022 |
| *Policy responses* | | |
| HF effect on 3-year mortality | −4.6pp (s.e. 1.7) | Costantini (2025), ≈300,000 VA veterans, IV |
| TF effect on long-run health | ≈ 0 | Costantini (2025). The outcome is mortality and health-service use; the model captures mortality and SUD status, not general morbidity |
| Housing stability, HF vs TF | sustained vs declining | HF RCT literature |
| PSH vs emergency shelter, property values | PSH raises, shelter lowers | shelter-siting hedonic literature |

**The Costantini result is the paper's largest single risk and belongs in the introduction.** The best available quasi-experimental evidence says the mortality benefit runs through *housing*, and that conditioning on treatment adds nothing detectable. The submitted paper concludes universal Treatment First dominates. The defensible position after the revision is the *generosity* result — expanding coverage dominates, conditionality is second-order and uncertain — which is

consistent with all of the above and is still a paper. Getting there requires reporting welfare and mortality as two dimensions, the AE's A9, rather than collapsing them into one ranking.

# 6 Diagnostics the model must pass

Two checks are run before any number is read as a result.

**Is life worth living?** Death is worth zero in (18). If any household has $V_j(s) < 0$ it strictly prefers to die, and for that household drug use is *rewarded* by the mortality it causes — inverting the paper's central mechanism for exactly the population the policy targets. The diagnostic reports the minimum $V$ over the benchmark distribution, the mass with $V < 0$, and whether they are disproportionately homeless, addicted or retired. If any mass appears, $\aleph$ is too small.

**The age profile of use, with and without $j_E$.** Referee 1's smaller comment 1 says the maximum experimentation age is in tension with the U-shaped age profile of use. The test is whether removing it produces a sensible profile or piles use onto the old, who have the least remaining life to lose. If use does concentrate at old ages, that is informative rather than embarrassing — it says the deterrent is the value of remaining life — but it has to be reported and explained rather than assumed away by a hard age limit.

# 7 Mechanisms: shutting channels and recalibrating $b$

## 7.1 Why this replaces the static appendix

Referee 1's comment 1(d) does not merely ask for intuition, it *specifies the design*: show that the mortality channel is the quantitatively important one by shutting channels off. Comment 1(e) says the current appendix is disorganised, static and overreaches. Referee 2 (C4) wants the drug-use policy function shown by income and wealth. Referee 3 (M2b) asks what patterns the Stone–Geary specification actually generates and where heterogeneity in $v$ comes from; (M2c) asks the sharpest theoretical question in any report — since drug use is rational and forward-looking, where do the externalities justifying intervention arise? The AE (A1) asks for a stylised section because the number of moving parts obscures the economics.

One exercise answers all five, run on the *calibrated* model rather than a separate static toy. That matters: a static model with no assets cannot address Referee 2's question about wealth, and a toy calibrated to nothing cannot establish that a channel is quantitatively important.

## 7.2 The design

Shut one channel at a time and, in each arm, **recalibrate $b_0$ so that drug participation returns to its benchmark value**. The recalibration is what makes the comparison informative: without it, removing the mortality penalty simply raises use everywhere and the level change swamps any compositional change. Holding participation fixed separates *sorting* from *level* and yields two statistics per arm:

| Statistic | What it measures |
|---|---|
| How far $b_0$ had to move | how much deterrence the channel was providing |
| The income gradient of use | *whom* it deterred |

The arms run in partial equilibrium at benchmark prices: the object of interest is the behavioural mechanism, not equilibrium feedback, and holding prices fixed isolates it as well as being far cheaper.

### 7.3 The arms

1. **Benchmark.** All channels on.
2. **No mortality penalty**, $\kappa_{\phi d} = 1$. The paper's headline claim.
3. **No employment penalty**, $\kappa_{ued} = 1$. Referee 1's rival hypothesis.
4. **No homelessness feedback**, $\kappa_{\phi H} = \kappa_{ueH} = \mu_H = 1$ and $\rho_H = 1$. Isolates the loop that makes homelessness and SUD mutually reinforcing.
5. **All private costs off at once.** Whatever deterrence survives is *resource diversion*: the budget constraint alone. Diversion cannot be switched off directly — drugs are priced inside a Cobb–Douglas bundle, so making them free sends $v$ to the corner — but it is measurable as the residual, which is what this arm reports.

### 7.4 What is reported

Table 7: Mechanism decomposition (structure; numbers pending the calibration)

| | Benchmark | No mortality penalty | No employment penalty | No homeless feedback | All private costs off |
|---|---|---|---|---|---|
| $b_0$ relative to benchmark | 1.00 | | | | |
| Participation (%, held fixed) | | | | | |
| SUD rate (%) | | | | | |
| Homelessness (%) | | | | | |
| Use rate, income Q1 (%) | | | | | |
| Use rate, income Q5 (%) | | | | | |
| User income / average | | | | | |
| experimenters only | | | | | |
| SUD only | | | | | |
| Users unemployed (%) | | | | | |
| Users homeless (%) | | | | | |
| Mean age of users | | | | | |
| Mean assets of users | | | | | |

### 7.5 The sharp prediction, and how it could fail

With preferences fixed across addiction status, becoming addicted no longer relaxes any constraint, so *all* deterrence must come from mortality, employment and the budget. If removing the mortality penalty requires only a small change in $b_0$ and leaves the income gradient nearly flat, the paper's central mechanism is not carrying the weight the text claims, and the honest response is to say so and re-centre on the generosity result. If $b_0$ must fall substantially and the gradient inverts, the mechanism is established numerically on the calibrated model — which is what Referee 1 asked for and what the static appendix could not deliver.

### 7.6 Where the externalities come from

Referee 3's M2c asks where the externalities justifying intervention arise, given that use is rational and forward-looking. They are of two kinds.

*Fiscal.* A household that becomes homeless or acquires SUD imposes $\chi_H$ and $\chi_D$ on the budget and reduces the tax base through lower earnings and higher transfer receipt. None is internalised, and all of it is measured directly by the model. This is the larger of the two and it is fully disciplined.

*Non-fiscal.* The disamenity term in (1). Referee 2 (C2) asked for it and said it was easy to introduce; it is, but it has no credible point estimate. Hedonic shelter-siting studies measure a *local, avoidable* disamenity — what people pay to live further away — and this model has no geography, so nobody can avoid anything and a capitalisation coefficient does not map onto an aggregate public bad. Corbae, Glover and Nattinger's externality is about neighbourhood housing *quality* rather than homelessness, and their identification comes from rent decontrol, so what transfers is their identification logic, not their functional form.

The recommendation is to set $\kappa_x = 0$ in the baseline, report a bounded sensitivity, and use the qualitative finding from the spatial literature — that permanent supportive housing *raises* nearby property values while temporary emergency shelter lowers them — as the substantive point. PSH is Housing First-like and transitional shelter is Treatment First-like, so the externality plausibly differs in *sign* between the two policies. That is a stronger argument than a made-up parameter. Note also that $\kappa_x = 0$ is a defensible baseline in a way $\kappa_{\text{cons}} = 0$ was not: that was the value at which the paper's conclusion reversed.

## 8 What each referee still wants

### 8.1 What has been done, and what it answers

Table 8: Model changes and the comments they answer

| # | Change | Answers |
|---|---|---|
| 1 | Homeownership, tenure, mortgages and negative assets removed | AE A4; R2 D1 |
| 2 | Housing-market GE removed | AE A5 |
| 3 | Drug price made an equilibrium object, $\psi_1$ normalised | R1 1(c); R3 M2d; AE A5 |

| # | Change | Answers |
|---|---|---|
| 4 | Cobb–Douglas weights fixed across SUD status | R2 C5; R3 M2a; AE A8 |
| 5 | $\kappa_{\text{cons}}$ deleted; homelessness externality in period utility, weighted by composition | R1 3(a), 3(b); R2 C2; R3 M2c; AE A9 |
| 6 | Endogenous treatment take-up, at a utility cost, with a take-up target | R1 1(b); AE A10 |
| 7 | Imperfect detection and false positives | R1 4(a), 4(b); AE A11 |
| 8 | Serious mental illness, six channels | R1 1(a), 2(c)(ii) |
| 9 | SUD wage penalty set to one in the baseline, with the evidence stated | AE A20 |
| 10 | UI drawn on entry, one year; four states collapse to three exactly | AE A2 |
| 11 | Estates to the government rather than lump-sum by type | AE A7 |
| 12 | Single addiction hazard; $\bar{v}$ and the 6.48 ratio gone | R3 M2b (partly) |
| 13 | Continuous labour supply from the FOC, welfare trap handled exactly | R3 m1; AE A3 |
| 14 | $\beta$ retargeted to wealth/GDP = 3 | AE A17 |
| 15 | Rent normalised; the whole housing grid calibrated | R2 D1 |
| 16 | Homeless SUD target corrected 37% $\rightarrow$ 21.7% | AE A16 |
| 17 | Drug-user income target split: users 95.1%, SUD 68.7% | R1 2(c)(i) |
| 18 | Housing burden targeted over all households, full CDF for validation | R1 2(a); AE A12 |
| 19 | Every parameter given a source or a target (§3) | AE A17 |
| 20 | Capital income taxed in the household budget; true median income | internal consistency |
| 21 | Mechanism ladder built | R1 1(d), 1(e); R2 C4; R3 M2b, M2c; AE A1 |

### 8.2 Referee 1 — outstanding

| # | Comment | Status |
|---|---|---|
| 1(d),(e) | Show the mortality channel is the quantitatively important one; reorganise the appendix | Ladder built, **not yet run** — needs the calibration |
| 2(a) | Model fit never reported | **Pending the calibration** |

| # | Comment | Status |
|---|---|---|
| 2(b) | Tables 3–4 misses framed too positively | **Writing**, once the misses are known |
| 2(c)(i) | Homeless 100% unemployed vs 40–52%; income $\approx 0$ vs 19.1% | Mechanism added; the employment share is **not yet a targeted moment** |
| 2(c)(ii) | Homeless too young | SMI added; **age distribution not yet computed**, and needs finer bins |
| 2(d) | Match policy-relevant RCT elasticities | **Second stage built, not yet run** |
| 2(e) | Price-elasticity moment not replicable | Model counterpart defined; **appendix not written** |
| 2(f) | Show identification numerically | **Not started.** One solve per parameter |
| 3(c) | Tone down; universal TF not unambiguously preferred | **Writing** |
| 3(d) | Solve proper transition paths | **Not built.** The critical path |
| 4(c) | Compliance, stigma and administrative costs of testing | **Not started.** Needs a real admin cost in the budget |
| 4(d) | Sensitivity ladder over addiction, recovery, mortality, employment | **Not started** |
| sm. 1 | Why a maximum experimentation age? | Now switchable; **diagnostic not yet run** |
| sm. 2–10 | Levels and percentage changes; typographical and citation fixes | **Writing** |

### 8.3 Referee 2 — outstanding

| # | Comment | Status |
|---|---|---|
| C1 | The elasticity of drug use to housing access | $\rho_H$ added and the moment defined; **not yet calibrated** |
| C2 | Homelessness externality | Implemented, $\kappa_x = 0$; **no credible calibration**, see §7.6 |
| C3 | Parental anxiety about children | **Open decision.** R2 says it "could be discussed"; a warm-glow term would answer this and AE A6 together |
| C4 | Show the drug-use policy function by income and wealth | **Not produced.** Free once calibrated |

| # | Comment | Status |
|---|---|---|
| C6 | Optimal voucher design; is $\chi_H$ too high? | **Not started.** One sweep of $\pi_V$ plus a $\chi_H$ sensitivity |

### 8.4 Referee 3 — outstanding

| # | Comment | Status |
|---|---|---|
| M1a | Stylised facts on drug use, voucher receipt and homelessness | **Not written** |
| M1b | Empirical basis for survival and unemployment hazards | Partly done in §3; $\bar{\phi}_j$ is still a placeholder and must come from life tables |
| M1c | Show the empirical income–drug-use relationship | Microdata extracted; **figure not made** |
| M2b | What patterns does Stone–Geary generate? | Ladder built, **not run** |
| M2c | Where do the externalities arise? | Enumerated in §7.6; the fiscal side needs quantifying from the ladder |
| m1 | Labour supply and the welfare cost of tax financing | Labour supply now continuous; **welfare decomposition not done** |
| m2 | Sensitivity as graphs, not tables | **Not started** |
| m3 | Vary voucher receipt continuously; Laffer curve? | **Not started** |

### 8.5 Associate Editor — outstanding

| # | Comment | Status |
|---|---|---|
| A1 | Stylised model section | Plan is to use the calibrated model plus the ladder; **not written** |
| A6 | Intergenerational language without intergenerational linkages | **Open decision**: delete the language, or add a warm-glow term (with R2 C3) |
| A9 | Report welfare and mortality as two dimensions | **Not done.** Requires the counterfactuals |
| A12 | Model fit; non-targeted moments | **Pending the calibration** |
| A13 | Key elasticities | **Second stage built, not run** |

| # | Comment | Status |
|---|---|---|
| A14 | Connect to Costantini, Cohen, Brounstein and Wieselthier, Abramson, Humphries et al. | **Writing** |
| A15 | Compare $\chi_H$ with Abramson and van Nieuwerburgh | **Not started** |
| A16 | Homelessness definition | Target corrected; the paper must still state that it targets *literal* homelessness and excludes doubling-up |
| A17 | Rewrite the calibration section | §3–4 are the raw material; **not yet in the paper** |
| A18 | Fewer counterfactuals in the body | **Writing** |
| A19 | "Addicted" → "suffering from substance use disorder" | **Writing**, global replace |

### 8.6 The critical path

Three things gate almost everything above.

1. **The first-stage calibration converging.** Unblocks R1 2(a), 2(b), 2(c), 1(d), R2 C4, R3 M2b, AE A12 at once.
2. **The transition-path solver.** Unblocks R1 3(d) *and* the whole of §4.2, which is R2 C1, R3 and AE A13 — what all three independently call the crux.
3. **The counterfactuals.** Unblock AE A9, A18, R2 C6, R3 m3, R1 3(c), 4(d).

Everything else on the outstanding lists is either writing or a single cheap run.